MILITARY SPECIFICATION

HOWITZER, LIGHT, T0WED, 105 MM, M204

This specification Is approved for use by US Army Research and Development Command, and is available for use by all Departments and Agencies of the Department Of Defense.

1. SCOPE

1.1 This specification covers one type of 105MM, light, towed, howitzer, M204 consisting of the M205 cannon, M46 recoil mechanism, M44 carriage and associated fire control.

2. APPLICABLE DOCUMENTS

2.1 The fallowing documents, *of the issue in effect on date of* invitation for bias or request for proposal, form a part of this specif *Ication* to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-H-5606	Hydraulic Fluid, petroleum base, Aircraft Missile and Ordnance
MIL-H-6083	Hydraulic Fluid, Petroleum Base, Preservation and Testing
MIL-H-6088	Heat Treatment of Aluminum Alloys
MIL-I-6866	Inspection, Penetrant, Method of
MIL-I-6868	Inspection process, Magnetic Particle
MIL-s-6872	Soldering Process, General Specif- cation for
MIL-H-6875	Heat Treatment of Steels (Aircraft Practice, Process for)
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-R-11468	Radiographic Inspections, Soundness Requirements for Arc and Gas Welds *In* Steel

Beneficial comments (recommendations, additions. deletions and any pertinent data which may be of use in improving this document shoud be addressed to: Commander. US Army .4rmament Research and Development Command, Attn. DRDAR-QA, Dover New Jersey 07801 by using the self-addressed standardization Document improvement Proposal (DD Form 1426) appearing at the cnd of this document or bylettir

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MIL-P-14232	Parts, Equipment and Tools for Army Materiel, packaging and Packing of
MIL-E-16053	- Electrodes, Welding, Bare Aluminum Alloys
MIL-W-45205	- Tungsten - Arc, Aluminum Alloys, Readily Weldable for Structures, Excluding Armor
MIL-I-45607	- Inspection Equipment, Acquisition Maintenance and Disposition of
MIL-R-45774	- Radiographic Inspection, Soundness Requirements for Fusion Welds in Aluminum and Magnesium Missle Components
MIL-P-45971	Product Engineering Quality Engineering System (PEQES) for Weapons and Support Materiel
MIL-C-45978	- Cannon, 105MM Howitzer, M205
MIL-A-48449	- Alignment Device, Optical, Gun Bore
MIL-T-48554	- Telescope, Panoramic: M137
MIL-Q-48555	- Quadrant, Fire Control: M17
MIL-Q-48556	- Quadrant, Fire Control: M18
MIL-M-48557	- Mount, Telescope and Quadrant: M171
MIL-T-48558	- Telescope, Elbow: M138
MIL-M-48559	- Mount, Telescope and Quadrant: M172

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-130	- Identification Marking of US Military Property
MIL-STD-171	- Finishing of Metal and Wood Surfaces
MIL-STD-248	- Welding and Brazing Procedure and Performance Qualification Performance Qualification
MIL-STD-1252	- Inertia Friction Welding Process, Procedure and Performance Qualification
MIL-STD-1261	- Welding Procedures for Constructional Steels
MIL-STD-1458	- Radioactive Materials: Marking and Labeling of Items, Packages and Shipping Containers for Identification A in Use, Storage and Transportation

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FEDERAL

Fed.Std. No. 595 Colors

DRAWINGS

ARRADCOM

F12010000 - Howitzer, Light, Towed, 105MM M204

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 other publications. - The following documents form a part of this specification to the extent specified herein. tinless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI 1346.1 - Surface Texture
ANSI Y14.5 - Dimensioning and Tolerancing

(APPlications for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017).

AMERICAN WELDING SOCIETY (AWS)

AWS A2.4 - Symbols for Welding and Mondestructive Testing

(Application for copies should be addressed to the American Welding Society, 2501 N.W. 7th Street, Miami, FL 33125).

NATIONAL BUREAU OF STANDARDS (NBS)

Handbook H28 - Screw - Thread Standards for Federal Services

(Appllcation for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402) .

3. REQUIREMENTS

3.1 First article.-Requirements for submission of a first article shall be specified in the contract (see 6.2). Unless otherwise specified, the first article shall include the pilot pack (see 3.1).

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3.2 Materials, construction and design.-The howitzer shall COnformed to the material, construction and design requirements specified herein and on Drawing J12010000.

3.2.1 Heat treatment.-Unless otherwise specified on the applicable drawings, heat treatment for aluminun alloys shall conform to MIL-H-6088 and for steel shall conform to MIL-H-6875.

3.2.2 Dimensions.-Dimensions and tolerances of parts and assemblies of the howitzer shall conform to the detailed drawings of Drawing F12010000. interpretation of the dimensions and tolerances shall be in accordance with ANSI Y14.5.

3.2.3 Surface texture.-Surface texture shall be as specified on the applicable drawings. Interpretation of surface roughness symbols shall be in-accordance with ANSI B46.1.

3.2.4 Protective finishes.

3.2.4.1 Inorganic finishes. - Parts requiring an inorganic finish shall be finished or coated as 'indicated on the applicable drawings and in accordance with MIL-STD-171.

3.2.4.2 Painting. -Parts and assemblies shall be painted as specified on the applcable drawings and in accordance with MIL-STD-171. The visible surfaces of attaching hardware (nuts, screws, washers, etc.) , exposed unpainted surfaces of mounting faces or blemished areas shall be primed and painted to match surrounding surfaces after final assembly. To avoid the necessity of painting exposed surfaces of cadmium plated hardware (including screws, nuts and washers), the hardware may be dyed with a supplementary chromate finish, Forest Green, Lusterless, of Spec. MIL-L-52909.

3.2.5 Welding.

3.2.5.1 General.-Interpretation of weld symbols specified on the applicable drawings shall be as defined in AWS A2.4. Actual production of weldments shall not proceed prior to Government approval of the contractor's welding procedures and test specimens of the applicable welding specifications. Qualification of arc welding procedures and welders shall be in accordance with MIL-STD-24.8. Qualification of inertia friction-welding shall be in accordance with MIL-STD-1252. All plug welds shall be flush and free of snags or protrusions. Grinding flush is not required unless specified on the drawing or in the welding specification. Weld splatter and similar excess weld shall be removed.

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3.2.5.2 Arc welding, aluminum alloy.-Welding of aluminum alloy parts shall conform to MIL-W-45205. Type 5356 welding electrode of MIL-E-16053 shall be used for 5000 series aluminum alloy welds. Type 5183 or 5556 welding electrodes of MIL-E_ 16053 may be used if Type 5356 welding electrode is unavailable. When welding aluminum alloy parts other than 5000 series aluminum alloy, the welding shall be in accordance with the specification referenced on the applicable drawings.

3.2.5.3 Arc welding, ferrous metals. -Welding of steel parts and assemblies shall be in accordance with weld classification of MIL-STD-1261 referenced on the applicable drawings.

3.2.5.4 Welding inertia. - Inertia friction welding shall be in accordance with MIL-sTD-1252.

3.2.5.5 Magnetic particle inspection. -When magnetic particle inspection is indicated on the drawing, it shall be in accordance with MIL-I-6868.

3.2.5.6 Penetrant inspection.-When penetrant inspection is indicated on the drawing, it shall be in accordance with MIL-I-6866.

3.2.5.7 Radiographic inspection.-Radiographic inspection shall be in accordance with MIL-STD-453. Aluminum inspection shall be to standards of MIL-R-45774 and steel inspection to standards of MIL-R-11468 as referenced on the applicable drawings.

3.2.6 Soldering.-Soldering of parts shall be as indicated on the applicable drawing and in accordance with MIL-S-6872.

3.2.7 Threads.- Threads shall conform to the thread series and class indicated on the applicable drawing and shall be in accordance with Handbook H28. Unless otherwise specified, coated or plated class 3 threads shall conform to the coated thread data specified in MIL-S-7742.

3.2.7.1 Torque, thread fasteners.-unless otherwise specified on the applicable drawing, threaded fasteners shall be torqued in accordance with the following torque requirements:

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Thread Size	Torque	
.112 - 48UNF	9 - 1 0	inch-pounds
.138 - 32uNF	16 - 18	inch-pounds
.164 - 32UNF	30 - 33	inch-pounds
.190 - 24uNF	40 - 45	inch-pounds
.190 - 32uNF	45 - 50	inch-pounds
.250 - 20UNF	8 - 1 0	foot-pounds
.375 - 16UNF	30 - 33	foot-pounds
.500 - 13uNF	75 - 80	foot-pounds
.625 - llUNF	150 - 165	foot-pounds

3.3 Performance and product characteristics.

3.3.1 Lunette assembly.-The lunette assembly shall be capable of being secured in the firing, down and towing position by the lunette operating lever.

3.3.2 Locking devices.-All locking and fastening devices for securing component parts shall be positive in action.

3.3.3 Handbrakes.-The handbrakes when engaged, shall prevent wheel rotation while the howitzer is in a level, ascending or descending travel position. When the handbrake is disengaged, the wheels shall turn freely with minimum drag.

3.3.4 Trunion adapter plates.-The trunion adapter plates shall maintain a verticle plane within + 0.2 mil when the cannon is elevated from O to 1275-mils. The trunion adapter plate keys shall be parallel with the cannon tube centerline within + 0.2 mil. These measurements shall be preceded by cross-leveling of-the carriage trunion centerline as specified in 3.3.9.1.

3.3.5 Wheel actuator.-The wheel actuator shall be capable of raising the wheels and lowering and raising the base Plate to the firing and travel positions. The actuator shall be self locking throughout the range of operation. The slip clutch limiter shall be adjusted to raise the weapon into travel lock with 400 + 50 lbs on the base plate and insure slippage with 550 $\pm$ 50 lbs.

3.3.6 Base plate.-The base plate socket in hte center of the base plate, is the point at which the weapon shall be secured and permit the weapon to be traversed 360 degrees. Stability shall be provided by grousers protruding from the bottom of the base plate.

3.3.7 Traversing mechanism.-With the howitzer in the firing position, the traversing mechanism shall be capable of traversing the howitzer 360 degrees in either direction. The torque required

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to accomplish traversing shall not exceed 115 inch-pounds at the handwheel shaft. The maximum backlash of the traversing handwheel shall not exceed 20 degrees rotation in either direction.

3.3.8 Elevating mechanism.-The elevating mechanism shall be capable of elevating and depressing the cannon tube from a poisition of minus 85 mils to positive 1333 mils. The torque required to elevate or depress at any point throughout the operating range shall not exceed 200 inch-pounds at the hand-wheel shaft. The maximum backlash of the elevating handwheel shall not exceed 20 degrees of rotation at any position in either direction of elevation or depression.

3.3.9 Fire Control.

3.3.9.1 Gun elevation - plumb travel.-The elevating trunnions of the carriage shall be cross leveled by having the cannon track a plumb line within 0.1 mil through a-range of O to 978 mils elevation minimum. The fire control instruments shall meet the requirements of 3.3.9.2 through 3.3.9.7. Testing shall be as specified in 4.6.9.2.

3.3.9.2 All knobs on the fire control instruments shall function properly without binding, looseness or excessive torque. All counters shall operate properly and all radioactive sources shall be illuminated, including those on the off-carriage equipment.

3.3.9.3 Alignment of mount M171, panoramic telescope M137 and quadrant M17.

a. The quadrant elevation level vial shall be adjusted, as necessary, to center the bubble when the cannon tube is at zero elevation and the elevation counter is set at 0000.

b. After boresighting on the target and setting the azimuth counter to 3200 mils, the panoramic telescope shall align on the alignment device at 4800 mils within ± 0.5 mil.

c. Elevate the cannon tube from O to 1275 mils in 200 mil steps. Relevel the M171 in pitch and cross-level for each 200 mil step. The verticle reticle line of the telescope shall not deviate from the test target verticle line by more than ± 0.6 mil from O to 899 mils elevation and not more than ± 1.5 mils from 900 to 1275 mils elevation.

3.3.9.4 Alignment of mount M172 and quadrant M18.

a. The elevation pads on the mount shall be parallel with the cannon tube within ± 0.2 mil.

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b. The quadrant elevation level vial shall be adjusted, as necessary, to center the bubble when the cannon tube is at zero elevation and the elevation counter is set at 0000.

3.3.9.5 Accuracy of elevation counters. Prior to proof firing, the elevation counter of the M17 and M18 quadrants shall reflect tube elevation and shall not exceed 1.0 mil deviation, excluding backlash, at any counter setting. After proof firing, the allowable deviation shall not exceed 1.5 mils, excluding backlash. The M18 cross level vial bubble shall remain centered within ± 1 mil (one full graduation) through all tube elevations.

3.3.9.6 Backlash.-Backlash shall not exceed 1.0 mil at any elevation counter reading.

3.3.9.7 Illumination.-The reticle markings, counter numerals, and level. vial bubble and graduation lines shall be clearly distinguishable when observed in ambient light conditions ranging from dusk into darkness.

3.3.10 Proof firing.-The howitzer shall withstand proof-firing as specified in Table I with no evidence of any system malfunction; There shall be no structural damage, permanent deformation or degradation of operating characteristics. Testing shall be as specified in 4.6.10.

TABLE I

No. of Rounds	Prop Charge	Zone	Elevation (mils) #10 mils	Proj
3	M67	1	0	Ml
3	M67	5	400*	Ml
3	XM200	8	800	Ml

3.3.11 Latch mechanism.-The latch mechanism shall retain the recoiling parts in battery position. When a force is applied to the lever/lanyard, the latch mechanism shall release the recoiling parts for forward run-up. Upon counter recoil the latch mechanism shall catch and retain recoiling parts in the battery position.

3.3.12 *Zone* setter.-The zone setter shall function such that the over travel requirements of Table II are met and the weapon latches.

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TABLE II

Nitrogen Pressure = 900 + 50 psi

Zone	QE (mils)	Over Travel (in.)
1	0	5 max.
8	1280	19 max.

3.3.13 Road test.-The howitzer shall withstand the following 50 mile road test without damaqe, permanent deformation or loosening of parts.

a. 35 miles, at 55 miles per hour, over hard surface roads.

b. 10 miles, at 15 miles per hour, over dirt roads.

c. 5 miles, at 10 miles per hour, over unimprofed or cross country roads.

3.3.14 Boresight retention.-AFter the proof-firing and road test, the line of sight of the panoramic telescope shall maintain its original setting in azimuth with respect-to the weapon tube within 0.3 mil. The direct fire telescope shall maintain its original setting with respect to the weapon tube within 0.8 mil in azimuth and 0.8 mil in elevation. Testing shall be as specified in 4.6.14.

3.3.15 Recoil mechanism.

3.3.15.1 General.-The recoil mechanism shall be filled with hydraulic fluid, bled and refilled until all air and measurable moisture are eliminated. The initial and subserquent filling of the hydraulic system shall be with hydraulic oil conforming to MIL-H-5606 or MIL-H-6083 and filtered through a 5 micron filter. The recoil mechanism shall conform to eht characteristics specified on Drawing F12009400 and the requirements of 3.3.15.2 through 3.3.15.4. Testing shall be as specified in 4.6.15.

3.3.15.2 Nitrogen leakage. When pressurized, the nitrogen filling end of the recuperator cylinder shall show no evidence of nitrogen leakage when observed for five minutes.

3.3.15.3 Foamed and contaminated oil.-After proof firing, oil from the recoil mechanism shall meet the following requirements.

a. Metallic and non-metallic particles shall not exceed 400 microns in size.

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h. Foam shall be 5 percent or less by volume within the first minute after withdrawal.

c. Aeration bubbles shall disappear when exposed to the atmosphere for 15 minutes.

3.3.15.4 Recoil oil reserve level.- Before and after proof firing, the oil level shall maintain the floating piston between the recuperator cylinder index lines. There shall be no evidence of oil leakage.

3.4 Cannon.-Cannon, 105MM Howitzer, M205 and the parts and assemblies comply with MIL-C-45978.

3.5 Identification and instructional marking.-Unless otherwise specified, agency identification marking,tire pressure . marking and-weight classification marking shall be applied in accordance with Table III. Other marking, including application of part numbers and instructional marking, shall be as specified on the applicable drawings and in accordance with MIL-STD-130. Serial numbers shall be furnished by the procuring activity.

TABLE III

Agency identification

Color: Lusterless white #37855
Location: Both sides of recuperator cylinder between rear and center yoke

Weight classification

Number: 2
Color: Lusterless black number (2) #37033 on a six inch diameter background or lusterless yellow #33538
Location: Left side of the box trail just to the rear of the traversing handwheel

Tire pressure

Color: Lusterless white #37875
Location: Any visible portion of the wheel support arm
Letters and Numerals: The marking will consist of numerals representing the proper tire pressure preceded by the letters TP
Size: Letters and numbers will be one inch

General data

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Markings shall be applied using stencil paint in accordance with TT-P-98 or pressure sensitive adhesive materials in accordance with MIL-M-43719.

A11 letters and numerals shall be of the gothic style and of the maximum practical size unless otherwise specified.

Colors shall be as specified in accordance with FED-STD-595.

3.6 Reliability, maintainability and durability.-Howitzers manufactured in accordance with this specification shall be free from any defects that compromise, limit or reduce the howitzer capability specified herein and may be subject to examination and test by the Government to prove howitzer capabilities and performance in accordance with the following:

3.6.1 Reliability.-The reliability requirement is a specified va ue of 2000 mean rounds between failure (MRBF) and a minimum acceptable value of 1200 MRBF as defined in AR 702.3.

3.6.2 Maintainability.-The maintainability requirement is 0.5 hours for mean time to repair (MTTR) at the organizational and 2.0 hours for direct support.

3.6.3 Durability.-The durability requirement is 15,000 Equivalent Full charge (EFC) rounds without rebuild or overhaul

3.7 Regulations pertaining to radioactive devices.-Marking, identification, handling and shipping of radioactive devices shall be in accordance with MIL-STD-1458.

3.8 Workmanship.-Workmanship and finish shall be in accordance with the highest practice used in manufacturing military weapons. Finished items and parts shall not exhibit poor material and processing such as seams, laps, laminations, cracks, visible steps, sharp edges, nicks, scratches? burrs, deformations and missing operations which may affect service-abillty, functioning, operation, appearance or safety. Fin and other extraneous metal shall be removed from cast or forged parts. Hammering to shape, salvage operations (including repair by welding) or other similar practices shall not be permitted without prior approval of the contracting efficer-

4. QUALITY ASSURANCE PROVISiONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract

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or purchase order, the supplier may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 The contractor shall perform, as a minimum, inspections in accordance with the specifications, quality assurance provisions, MIL-P-45971 and the contract. These minimum inspections shall not be construed as relieving the contractor of his responsibilities under terms of the contract to furnish the Government with items complying with and conforming to the requirements of the contract, drawings and specifications.

4.2 Quality assurance terms and definitions.-Quality assurance terms and definitions used herein are in accordance with M1L-STD-109,

4.3 Classification of inspection.-The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

4.4 First article inspection.-The first article shall be submitted for inspection in accordance with the contract. The first article sample shall be representative of the production process to be used during quantity production. The first article shall be subjected to the quality conformance inspection specified herein and such other inspections as necessary to determine that all requirements of the contract have been met.

4.5 Quality conformance inspection.

4.5.1 Inspection lot.-Unless otherwise specified herein, the formation, size and presentation of inspection lots of howitzers and parts shall be in accordance with MIL-STD-105. Howitzers shall be assembled from lots of component parts that meet all the inspection requirements specified herein.

4.5.2 Examination.

4.5.2.1 Parts and assemblies.-Parts and assemblies shall be examined to determine compliance with the requirements of 3.2.1 through 3.2.7.1, 3.5, 3.8 and MIL-P-45971. Parts and assemblies failing to comply shall be rejected.

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4. 5.2.2 Howitzers.-Visually and manually examine each howitzer to determine conformance with the applicable requirements. Each step in the examination shall include a visual examination for proper cleaning, the presence of the specified protective coating and to determine the general quality, completeness of manufacture, asembly and workmanship (see 3.8). Visually examine all markings to assure that they are correct and legible.

4.5.2.3 Packaging.-Unless otherwise specified (see 6.2) inspection to determine compliance with preservation, packing and marking requirements of the applicable packaging documentation, for the level designated in the contract, shall be as specified in MIL-P-14232.

4.5.2.4 Quality Inspection Plan.-The contractor shall prepare and submit to the Government for review and approval a Quality ,Inspection Plan. The plan shall identify the component/ assembly drawing number, describe the defect characteristic, identify the inspection/test equipment, specify the sample size and accept/reject numbers. Submittal of the plan shall be to Commamder, ARRADCOM, Attn: DRDAR-QAR-R, Dover, NJ 07801.

4.5.3 Testing.

4.5.3.1 Failure data.-Unless otherwise specified herein, all tests shall be conducted on a complete howitzer. If test requirements cited herein are not complied with, acceptance of the howitzer shall be deferred and the contractor shall accomplish, as applicable, the following actions:

a. Conduct a failure analyses study performing a dimensional, physical and visual examination of the components which are suspected to be the cause of failure or malfunction.

b. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect(s) in future production.

c* Examine howitzers, partially assembled howitzers, and components (including components and subassemblies at in-process or final assembly) to assure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.

d. Submit the results of the failure analysis and the corrective actions taken to the Government for review and approval prior to submitting a reconditioned lot or reconditioned howitzers for retest.

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4.5.3.2 Performance and product characteristic testing.- Each completely assembled howitzer shall be tested for compliance with the requirements of 3.3.1 through 3.3.15.4 using the " test methods specified in 4.6.1 through 4.6.15.2. Failure to comply with the requirements shall be cause for rejection.

4.5.3.3 Proof firing testing

4.5.3.3.1 Unless otherwise specified (see 6.2), the first twenty consecutive production howitzers shall be subjected to proof firing using the test method specified in 4.6.10. Failure of a howitzer to comply with the requirements of 3.3.10 shall be cause for rejection. If a howitzer fails to comply with the specified requirement, proof firing of each production howitzer shall be performed beginning with the next unit produced after rejection and continuing until twenty consecutive production howitzers comply with the specified requirement.

4.5.3.3.2 After twenty consecutive production howitzers comply with the specified requirement and if the production process is unchanged, one howitzer from" every twenty produced thereafter shall be randomly selected and subjected to the proof firing test. If a howitzer fails to comply with the specified requirement, it shall be rejected and the sample plan shall revert back to the requirement specified in 4.5.3.3.1. Also, if the production process changes, the sample plan shall revert back to the requirement specified in 4.5.3.3.1.

4.5.3.4 Latch mechanism testing.-Each howitzer subjected to proof firing testing shall be examined for compliance with the requirement of 3.3.11. Failure of the howitzer to comply with the specified requirement shall be cause for rejection. Testing shall be as specified in 4.6.11.

4.5.3.5 Zone setter testing.-Each howitzer subjected to proof firing testing of 4.5.3.3 shall be examined for compliance with the requirement of 3.3.12. Failure of the howitzer to comply with the specified requirement shall be cause for rejection. Testing shall be as specified in 4.6.12.

4.5.3.6 Road testing.-Each howitzer subjected to proof firing testing of 4.5.3.3 shall be road tested to determine compliance with the requirement of 3.3.13. Failure of the howitzer to comply with the specified requirement shall be cause for rejection. Testing shall be as specified in 4.6.13.

4.5.3.7 Boresight retention testing.-After proof firing and road test boresight tested to determine compliance with the requirement of 3.3.14. Failure of the howitzer to comply with the specified requirement shall be cause for rejection. Testing shall be as specified in 4.6.14.

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4. 5.3.8 Recoil mechanism testing.-Each howitzer subjected to proof firing testing of 4.5.3.3 shall be examined for compliance with the requirements of 3.3.15.1 through 3.3.15.4. Failure of the howitzer to complY with the specified requirements shall be cause for rejection. Testing shall be as specified in 4.6.15.

4.5.4 Inspection equipment.-Unless otherwise specified herein, responsibility for acquisition, calibration, maintenance and disposition of acceptance inspection and test equipment on List of Inspection Equipment Number EL 12010000-I and for all other inspection equipment required by applicable specifications, shall be in accordance with MIL-I-45607.

4.6 Test methods

4.6.1 Lunette assembly test.-Disengage the spring detent by pulling out the lunette operating lever knob and rotating the lever to the down position. After moving the lunette to the tow position, rotate the lunette operating lever to the rear until the spring detent locks the lunette in place. Repeat the procedure for placing the lunette in the down and firing position.

4.6.2 Locking device test.-All locking devices shall be tested for secure assembly and positive action.

4.6.3 Handbrake test.-The howitzer, in the travel lock and towing position, shall be placed on a 15 degree slope with the handbrakes applied. The surface texture of the slope shall. be such that the tires will not slip. When the hand-brakes are disengaged, the tires shall rotate freely. The requirement may also be checked by applying an 800 $\pm$ 80 pound push or pull force on the lunette with the weapon in the travel position on a level surface, or by applying a 500 $\pm$ 50 pound-foot torque to each wheel.

4.6.4 Trunion adapter plate test.-A calibrated gunner's quadrant and trunion adapter plate inspection equipment shall be used to determine trunion cross-level and trunion adapter plate to cannon tube parallelism.

4.6.5 Wheel actuator test.-With the howitzer in the travel lock and towing position, release both handbrakes. Disengage the suspension travel lock by pushing down the travel lock release lever. If the travel lock does not move, turn the actuator crank counter 'clockwise to relieve the pressure on the travel lock.

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While holding down the travel lock release lever, depress the safety lever and turn the actuator clutch release clockwise and allow the base plate 'to fall freely to the ground in not less than 3 seconds and not more than 10 seconds. As soon as the base plate touches the ground, return the safety lever to its original position. Rotate the actuator crank clockwise until the wheels retract. After retracting the wheels, rotate the actuator crank counterclockwise until the wheels are fully extended and the travel lock engages.

4.6.6 Base plate test.-With the base plate lowered to the ground and secured by the pivot and clip, the howitzer shall then be traversed 360 degrees in either direction by rotating the traversing handwheel.

4.6.7 Traversing mechanismtest.-With the howitzer emplaced, carriage level and tube at zero degrees elevation; the weapon shall be traversed 360 degrees in both directions by rotating the traversing handwheel. While traversing, torque and backlash measurements shall be taken at zero degrees, + 15 degrees and + 22 degrees. A torque wrench graduated in inch-pound shall be used for measuring loads at the shaft of the handwheel.

4.6.8 Elevating mechanism test.-With the howitzer emplaced and carriage level the tube shall be elevated and depressed through -4 degrees to +70 degrees elevation by rotating the elevating handwheel. While elevating and depressing, torque measurements shall be taken at -4 degrees, +5 degrees, +10 degreesm +15 degrees, +25 degrees, +35 degrees, +45 degrees, +55 degrees +65 degrees and +70 degrees. A torque wrench graduated in inch-pound shall be used for measuring loads at the shaft of the handwheel upon elevation and depression. Clockwise and counter-clockwise measurements shall be taken.

4.6.9 Fire control test.

4.6.9.1 Fire control test equipment and orientation of fire control instruments.

a. Mechanical jacks with a minimum of one square foot base for leveling the carriage trunions.

b. A plumb line approximately 1/16 inch in diameter and about 45 feet long shall be suspended one-foot or less from the end of the muzzle. The plumb line shall be damped.

c. .A boresight target shall be constructed on non-warping and nonshrinking material and placed approximately 50 meters from the fire control instruments. The test target pattern (Figure 1)

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shall be used for fire control and cannon bore alignment. The target shall be placed at the specified distance in a plane perpendicular to the cannon bore. The vertical line on the target shall be set plumb with the plumb line established for cannon tube plumb travel. The horizontal target line shall be established by the zero elevation setting of the cannon tube which will establish the required height of the target from the ground.

d. The borescope used for cannon tube elevation plumb travel shall have a cross hair reticle mounted in a self-aligning adapter to be inserted into the breech of the cannon so that the reticle has a focusing range of five meters to infinity. The reticle of the telescope shall be collimated with respect to the outside diameter of the adapter within 0.1 mil. The borescope shall have a magnification of four or greater.

e. Peep sights having a aperture of 1/16 inch may be used to eliminate parallax in the panaramic telescope and elbow telescope, as necessary.

The carriage trunion shall be cross-leveled as a preparatory step for cannon tube plumb travel.

g. Use cannon tube leveling pads for zeroing out cannon tube. Only a calibrated gunner's quadrant shall be used as an inspection aid on cannon tube pads.'

h. Sight through the borescope and align the Target line A and B (Figure I) in coincidence with the borescope reticle lines.

i. Set the level vials on the mount M171 and quadrant M17 to the level position (bubbles centered) .

j Sight through the panoramic telescope and align the reticle to the intersecting target lines C and D (Figure I) with the elevation and azimuth knobs. Ascertain that the level vial bubbles for pitch and cross-level are centered and the azimuth counter reads 3200.

k. Set the reticle of the telescope M138 to the zero position and align to the target intersecting lines E and F (Figure I). For reticle/target line coincidence, make all adjustments with the boresight screws on the M172 mount.

1. The quadrant M18 level vial bubble shall be centered with the elevation reading at 0000. The cross-level vial bubble shall be centered when boresight adjustments are set.

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Counter backlash may be eliminated *by* always making the last movement in the elevation counter mechanism from high to low.

n. Backlash in the panoramic telescope M137 may be eliminated by always approaching the target from left to right.

4.6.9.2 Gun elevation-plumb travel.-The howitzer shall be cross leveled using mechanical jacks. Using a borescope, track a plumb line through a range of O to 978 mils (min.) elevation.

4.6.9.3 Alignment of mount M171, panoramic telescope M137 and quadrant M17

a. Set the cannon tube at zero elevation and center all level vial bubbles.

b. The quadrant elevation level vial shall be adjusted as necessary and shall meet the requirements of 3.3.9.3.a.

c. Subsequent to boresighting, the panoramic *telescope* shall align on the M139 alignment device at 4800 mils within the limits specified in 3.3.9.3.b. Remove the alignment device before elevating the gun.

d. The gun tube shall be elevated from zero to 1275 mils in 200 mil increments. The pitch and cross-level knobs shall be rotated to relevel the mount and the level vial bubbles centered. Caution should be taken when rotating the knobs in one direction so as not to introduce backlash. If override should occur, repeat the procedure with the applicable knob. At each 200 mil increment the level vials shall be centered and the verticle line of the telescope's reticle shall not deviate more than the limits specified in 3.3.9.3.c throughout the range of zero to 899 mils. Continue the above procedure starting at 900 mils to 1275 mils in 200 mil increments. For the higher elevations, deviation shall not exceed the limits specified in 3.3.9.3.C.

4.6.9.4 Alignment of telescope M138, mount M172 and guandrant M18

a. The cannon tube shall be set at zero elevation. Place a calibrated gunner's quadrant on the pads provided on the mount. The elevation pads shall be parallel to the cannon tube within the limits specified in 3.3.9.4.a.

b. Set the quadrant to read 0000. The elevation level vial bubble shall be centered within the limits specified in 3.3.9.4.b, as measured with the calibrated gunner's quadrant.

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4.6.9.5 Quadrants M17/M18 elevation counter accuracy.-Prior to proof firing, the elevation counters' accuracy shall not exceed the deviation limit specified in 3.3.9.5, excluding backlash at any cannon tube elevation setting. Subsewuent to proof firing. the deviation shall not exceed maximum limit specified in 3.3.9.5, excluding backlash.

4.6.9.6 Backlash-quadrants M17/M18.-The elevation knobs on each quadrant shall be tested for maximum allowable backlash at any elevation setting. Backlash tests shall be taken in clockwise and counterclockwise directions. Backlash measurements shall be taken using a calibrated gunner's quadrant. Backlash shall not exceed the limit specified in 3.3.9.6.

4.6.9.7 Operability and illumination.-Mounted fore control shall have a visual and tactile examination for knob function to detect looseness, binding or excessive torque. Counters shall operate" properly and the numerals shall be legible and readable at light levels ranging from dusk into darkness. Level vial bubbles and graduation lines shall be clear and discernible at light levels ranging from dusk into darkness. Radioactive illuminated retitles in the panoramic and elbow telescopes shall have clear and concise line patterns that are discernible at light levels ranging from dusk into darkness. Off carriage fire control shall be examined in the same manner as indicated above.

4.6.10 Proof firing test.-The howitzer shall be emplaced, layed and fired in accordance with the operating procedures contained in PDTM 9-1015-248-10. The projectile, charge zone and elevation shall be in accordance with Table I.

4.6.11 Latch mechanism test.-During the proof firing test, observation shall be made for compliance with the requirements specified in 3.3.11.

4.6.12 Zone setter test.-During the proof firing test, measurements shall be made to determine compliance with the requirements specified in Table 11 of 3.3.12.

4.6.13 Road test.-The howitzer shall be prepared for travel and engaged to a prime mover in accordance with TM9-1015-248-10. Observation shall be made for compliance with the requirements specified in 3.3.13.

4.6.14 Boresight retention.-All lines of sight of the optical instruments, including borescope, shall be secured in coincidence with their respective arming points as defined in 4.6.9.1. The exact position of the carriage shall be marked to

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insure repositioning after performing the proof firing and road test to determine possible displacement of any lines of sight. Upon completion of the proof firing and road test, the carriage shall be positioned over the original marked location and the reticle of the borescope shall be placed in coincidence with its respective target aiming point (lines A and B of Figure I). Observe the respective lines of sight to determine their relationship to the original settings. The panormic telescope and direct fire telescope lines of sight shall not have deviated more than the tolerance specified in 3.3.14.

4.6.15 Recoil mechanism test.-The following tests shall be performed prior and subsequent to the proof firing test.

4.6.15.1 Nitrogen leakage.-A solution of soap and water shall be applied to the nitrogen filling end of the pressurized recuperator cylinder. There shall be no evidence of soap bubbles when the cylinder is observed for a period of five minutes.

4.6.15.2 Foamed and contaminated oil.-Oil shall be withdrawn slowly from the recoil mechanism into a glass beaker. After exposure to the atmosphere for one minute, the quantity of foam shall not exceed five percent by volume and after fifteen minutes, there shall be no operation bubbles. After exposure to the atmosphere for one hour, the oil shall be examined for contamination. Oil that contains sediment at the bottom of the beaker is contaminated.

4.6.16 Reliability, maintainability, availability and durability demonstration.-Demonstration shall be conducted by the Government in accordance with a Government prepared demonstration test plan. The test shall be for information only and a howitzer exhibiting a failure shall be cause for the contractor to perform a failure analyses. The Government shall verify the results of the failure analysis and assure that deficiencies causing the failure are corrected on all weapons and components in process at the contractor's plant.

5. Packaging.

5.1 Pilot pack.-A pilot pack shall consist of a complete howitzer preserved in accordance with Packaging Data Sheet for the level of protection specified in the contract (see 6.2), packed Level B and forwarded as specified in 6.2.

5.2 Levels A and B.-preservation, packing and marking shall be in accordance with packaging Data Sheet J12010000 for the level of protection specified in the contract (see 6.2).

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5.3 Repair parts.-Repair parts shall br preserved, packed and marked in accordance with the Packaging Data Sheet or other packaging requirements, as specified in the contract (see 6.2).

6. NOTES

6.1 Intended use.-The M204 Howitzer is a lightweight, towed, 105MM field artillery weapon suited for close support operations. This weapon is highly mobile and is capable of transversing 360 degrees with an elevation range of minus 85 to plus 1335 mils. The weapon is capable of being air-lifted and transportable by helicopter.

6.2 ordering data.-Procurement documents should specify the following:

a. Title, number and date of this specification.

b. List of drawings, specifications and publications pertinent to the howitzer, showing applicable revision dates.

c. Shipping instructions for first article and pilot pack.

d. Marking, if different, and block of serial numbers.

e. Number of howitzers required for proof firing, if other than specified.

f. Packaging inspection, if different.

g. Selection of applicable level of protection A or B.

h. Packaging requirements for repair parts.

Custodian:
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Preparing Activity:
Army-AR

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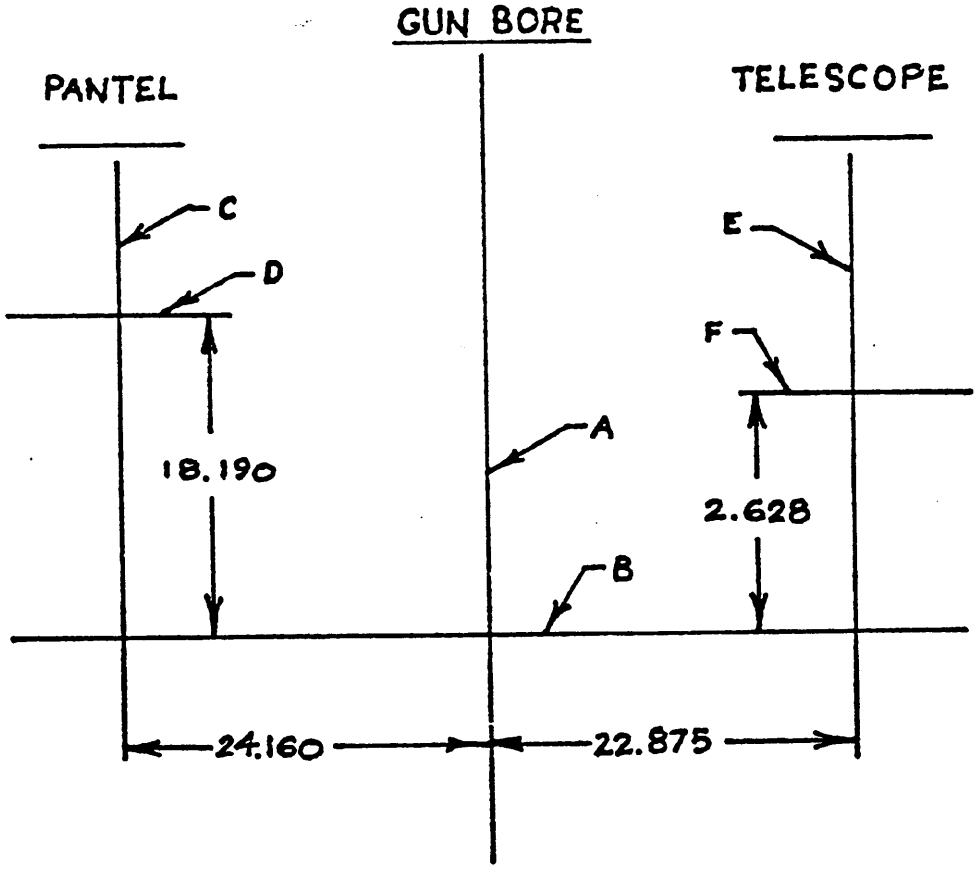


Figure I

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-H-63192 Howitzer, Light, Towed, 105MM, M204

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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